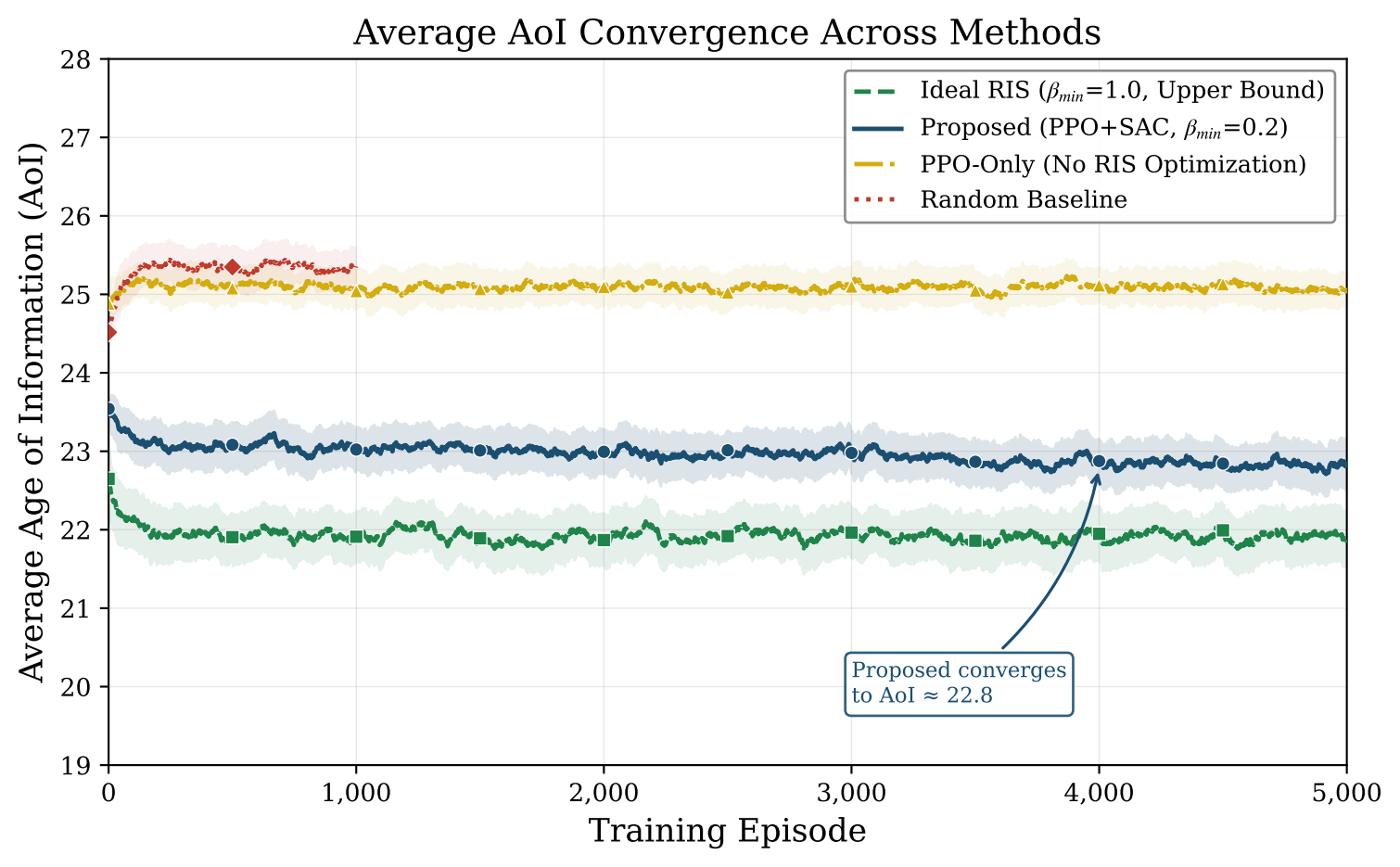

Average AoI Convergence Across Methods
Ideal RIS (β_{min}=1.0, Upper Bound)
Proposed (PPO+SAC, β_{min}=0.2)
PPO-Only (No RIS Optimization)
Random Baseline
Proposed converges to AoI ≈ 22.8
Average Age of Information (AoI)
Training Episode
28
27
26
25
24
23
22
21
20
19
0
1,000
2,000
3,000
4,000
5,000